Example 7 Simplify : $\frac{3}{7}+\frac{2}{3}$

Solution By Cross Multiplication technique, $\frac{3}{7}\times\frac{2}{3}=\frac{(3\times3)+(2\times7)}{7\times3}=\frac{9+14}{21}=\frac{23}{21}$.

Think about the situation

Vani has $\frac{3}{4}$ litre of water in her bottle. She drank $\frac{1}{2}$ litre of it. How much water is left in the bottle? To find the amount of water remaining in her bottle, the amount of water consumed by her must be subtracted from the amount of water she had initially. That is $\frac{3}{4}-\frac{1}{2}$. This is solved in the following example.

Example 8 Simplify: $\frac{3}{4}-\frac{1}{2}$

Solution Common multiple of 2 and 4 is 4

Equivalent fraction of $\frac{1}{2}$ is

$$\frac{1}{2}=\frac{1\times2}{2\times2}=\frac{2}{4}$$

Now, $\frac{3}{4}-\frac{1}{2}=\frac{3}{4}-\frac{2}{4}=\frac{3-2}{4}=\frac{1}{4}$

Therefore, Vani has $\frac{1}{4}$ amount of water in the bottle. This can be verified by the following diagram.



$\frac{3}{4}$ – $\frac{2}{4}$ = $\frac{1}{4}$

Example 9 Find the difference between $\frac{3}{4}$ and $\frac{2}{7}$.

Solution To find the difference, first we should know, which is bigger between the two given fractions ? How do we find out ?

We know that comparing 'like fractions' is easy, but these are unlike fractions. So, let us convert them into 'like fractions' and compare. Again, we use common multiple 28 (4×7) to get equivalent fractions of $\frac{3}{4}$ and $\frac{2}{7}$ as $\frac{21}{28}$ and $\frac{8}{28}$. Here, $\frac{21}{28}>\frac{8}{28}$ and hence $\frac{3}{4}>\frac{2}{7}$.

Therfore, $\frac{3}{4}-\frac{2}{7}=\frac{21}{28}-\frac{8}{28}=\frac{13}{28}$.

This can also be done by Cross Multiplication technique as $\frac{3}{4}\times\frac{2}{7}=\frac{(3\times7)-(2\times4)}{4\times7}=\frac{21-8}{28}=\frac{13}{28}$.

Try these

i) $\frac{2}{3}+\frac{5}{7}$ ii) $\frac{3}{5}-\frac{3}{8}$

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